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Authorship Identification using Applied Machine Learning

Abstract

Authorship analysis is a well-established research area in natural language processing that aims to identify, verify, or profile authors based on their writing style. This project presents an applied machine learning system for authorship identification that aligns closely with the stylometric principles and machine learning techniques surveyed in *A Survey on Authorship Analysis Tasks and Techniques* by Misini et al. (2022). Building upon the surveyed literature, the proposed system focuses on authorship attribution using lexical stylometric features and classical machine learning methods, while extending prior work by explicitly handling AI-generated text and open-set authorship scenarios. The system combines TF-IDF-weighted character and word n-grams with a linear Support Vector Machine (SVM), augmented by confidence-based rejection and similarity-based new author registration.

1. Introduction

Authorship Analysis (AA) aims to determine characteristics of an author or identify the author of a text using measurable linguistic features. As summarized by Misini et al. (2022), AA comprises three main subtasks: authorship attribution, authorship verification, and authorship profiling. This project is positioned specifically within the **authorship attribution** subtask, where the goal is to identify the author of an anonymous text from a set of candidate authors.

Recent advances in text generation by large language models have introduced new challenges to authorship analysis. Traditional AA systems assume that all texts originate from human authors, an assumption that no longer holds in modern digital environments. This project addresses this gap by integrating AI-generated text into the authorship attribution framework.

2. Alignment with the Surveyed Research

The survey by Misini et al. provides a comprehensive overview of authorship analysis techniques, emphasizing three core components:

1. Stylometric feature extraction
2. Machine learning–based classification
3. Task-dependent system design

This project directly follows these principles:

- It adopts **stylometry** as the foundation for representing writing style.
- It uses **machine learning classifiers**, particularly SVMs, which the survey identifies as one of the most consistently effective methods in authorship attribution tasks.
- It treats authorship attribution as a **multi-class classification problem**, consistent with the majority of surveyed studies.

Thus, the theoretical foundation and methodological direction of this project are strongly aligned with the reviewed literature [filecite]turn2file0[.

3. Problem Formulation

The task is formulated as an authorship attribution problem with additional real-world constraints:

- Given a text document, identify its author from a set of known authors.
- If the author is not present in the known set, the system should reject the attribution.
- AI-generated texts should be identifiable within the same framework.

Unlike many studies surveyed by Misini et al., which assume a closed-set of candidate authors, this project explicitly models **open-set authorship**, allowing the system to express uncertainty.

4. Datasets

4.1 Human Authorship Dataset

The Blog Authorship Corpus was used as the primary human-authored dataset. This dataset aligns with the survey’s observation that blogs, articles, and informal digital texts are increasingly used in modern AA research. From this corpus, 40 authors were selected, each contributing an equal number of samples to ensure balanced learning.

4.2 AI Authorship Dataset

To incorporate AI-generated text, a Kaggle dataset containing AI-written essays was used. Instead of framing AI detection as a binary classification task, AI texts were grouped into 10 pseudo-authors. This approach is consistent with the survey’s emphasis on similarity-based and attribution-based formulations of authorship-related tasks.

5. Feature Engineering

The survey highlights feature engineering as the most critical factor in authorship attribution. In particular, lexical features and n-gram representations are identified as among the most effective stylometric markers.

5.1 Lexical Stylometric Features

This project employs:

- **Character n-grams (3–5):** Capture low-level stylistic habits such as punctuation, spacing, and morphological patterns.
- **Word n-grams (1–2):** Capture habitual phrasing and short-range lexical patterns.

These features fall under the **lexical feature category** described in the survey and are widely reported as effective across languages and text types [filecite?turn2file0?].

5.2 TF–IDF Weighting

Term Frequency–Inverse Document Frequency (TF–IDF) weighting is applied to emphasize distinctive stylistic patterns while suppressing common language usage. This aligns with the survey’s discussion on weighting schemes used in stylometric analysis.

5.3 Feature Fusion

Character and word n-gram features are combined at the feature level to form a high-dimensional sparse representation. This reflects the survey’s observation that combining multiple feature categories often yields improved performance.

6. Classification Method

A linear Support Vector Machine (SVM) is used as the primary classifier. The survey notes that SVMs are among the most frequently used and best-performing classifiers in authorship attribution tasks, particularly when combined with lexical features [filecite?turn2file0?].

The choice of a linear SVM is motivated by:

- Its effectiveness in high-dimensional sparse feature spaces
 - Its robustness in multi-class stylometric classification
 - Its computational efficiency compared to more complex models
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7. Open-Set Authorship Handling

While the survey primarily focuses on closed-set authorship attribution, it also discusses authorship verification and similarity-based approaches. Building on this, the proposed system introduces confidence thresholding on SVM decision scores. Predictions with low confidence are labeled as UNKNOWN_AUTHOR, preventing forced attribution.

This design extends traditional authorship attribution systems by incorporating uncertainty awareness.

8. New Author Registration

For texts identified as belonging to unknown authors, the system allows registration of new authors using similarity-based matching. Cosine similarity between TF-IDF feature vectors is used to compare new texts with stored author representations.

This approach is conceptually related to the instance-based and similarity-based strategies discussed in the survey, but is applied here as a practical system extension rather than a standalone task.

9. Key Differences from the Surveyed Research

While the project aligns strongly with the surveyed literature, it differs in several important ways:

1. **Integration of AI-generated text:** The survey primarily addresses human authorship, whereas this project explicitly incorporates AI as author classes.
2. **Open-set formulation:** Many surveyed studies assume a fixed set of authors; this project allows for unknown authors.
3. **System-level design:** The project emphasizes deployment-oriented features such as model persistence, user interaction, and incremental author registration.

4. **Applied focus:** Rather than proposing new features or models, the project demonstrates how established techniques can be combined into a realistic applied ML system.
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10. Limitations

Consistent with the survey's conclusions, this project acknowledges several limitations:

- Short texts provide insufficient stylometric signal
- Authors may exhibit style variation across contexts
- AI writing styles may evolve over time

These limitations are inherent to stylometric approaches and are not specific to the proposed style.

11. Conclusion

This project demonstrates an applied machine learning system for authorship identification that is strongly grounded in established stylometric research. By following the feature extraction strategies and classification approaches surveyed by Misini et al. (2022) and extending them to handle AI-generated text and open-set authorship, the system bridges the gap between academic research and realistic deployment scenarios. The work highlights that careful feature engineering, appropriate model selection, and uncertainty-aware design are central to effective authorship analysis.

Reference

Misini, A., Kadriu, A., & Canhasi, E. (2022). *A Survey on Authorship Analysis Tasks and Techniques*. SEEU Review, 17(2).